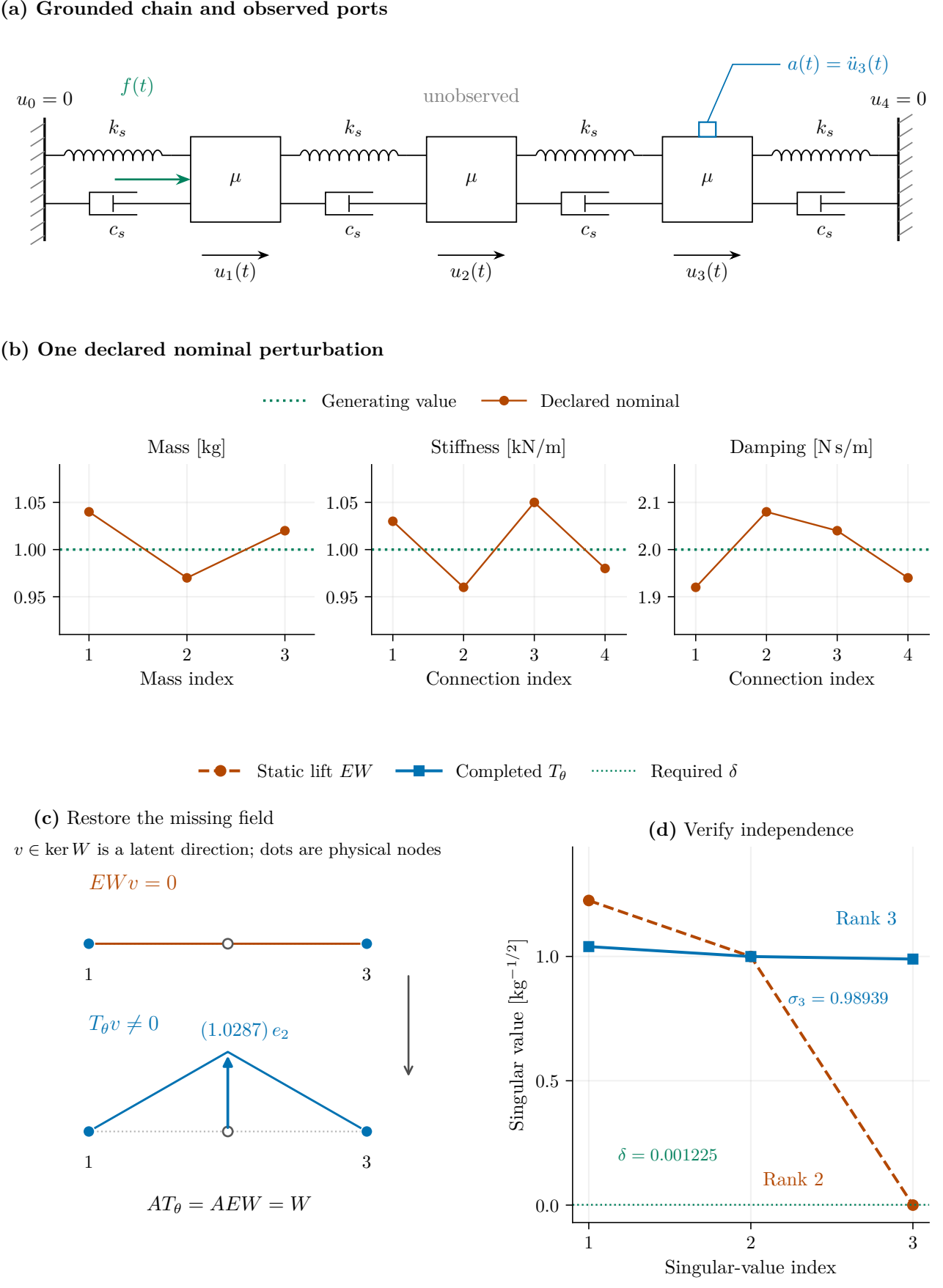

(a) Grounded chain and observed ports
$u_0 = 0$
$f(t)$
unobserved
$a(t) = \ddot{u}_3(t)$
$u_4 = 0$
k_s
μ
c_s
$u_1(t)$
$u_2(t)$
$u_3(t)$
(b) One declared nominal perturbation
Generating value
Declared nominal
Mass [kg]
Stiffness [kN/m]
Damping [N s/m]
Mass index
Connection index
Static lift EW
Completed T_θ
Required δ
(c) Restore the missing field
$v \in \ker W$ is a latent direction; dots are physical nodes
$EWv = 0$
$T_\theta v \neq 0$
$(1.0287)\, e_2$
$AT_\theta = AEW = W$
(d) Verify independence
Singular value [kg$^{-1/2}$]
Singular-value index
Rank 3
$\sigma_3 = 0.98939$
$\delta = 0.001225$
Rank 2